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# The Confluence of Digital Twin and Blockchan Technologies in Industry 5.0: Transforming Supply Chain Management for Innovation and Sustainability

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## Abstract

In the era of Industry 5.0, characterized by the seamless collaboration between humans and machines, the integration of digital twin technology (DTT) and blockchain technology (BCT) is poised to revolutionize supply chain management. This research explores the impact of DTT and BCT on achieving sustainable and efficient supply chain operations. Digital twins, virtual replicas of physical systems, enable real-time analysis and simulations, enhancing decision-making and operational efficiency. Simultaneously, blockchain technology ensures transparency and security in supply chains by maintaining unchangeable transaction records. This paper delves into the advantages and challenges presented by these technologies, examining real-world case studies across various industries. The study reveals that the combination of DTT and BCT creates a symbiotic relationship, driving continuous monitoring and validation of supply chain processes. This integration aligns with global sustainability goals, emphasizing resource optimization and waste reduction. However, data privacy, scalability, and interoperability remain significant barriers. A comprehensive approach is advocated to overcome these challenges, emphasizing ethical and environmental norms. Furthermore, this research offers insights into the mediating role of sustainable supply chain management (SSCM) practices and dynamic capabilities in the relationship between Industry 5.0 technologies and operational resource utilization (ORU) performance. It highlights the need for a sophisticated strategy in implementing technology adoption initiatives. This study contributes to theoretical advancements in Industry 5.0 by elucidating the complex interactions between DTT, BCT, and SSCM, paving the way for future research. Additionally, it provides valuable policy implications, guiding policymakers to prioritize innovation, transparency, and sustainability in the industrial sector. Integrating DTT and BCT can reshape supply chain dynamics, fostering a future marked by efficiency, innovation, and environmental responsibility.

**Keywords** Digital twin technology · Blockchain technology · Sustainable supply chain management · Resource utilization · Industry 5.0

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## Introduction

In the era of Industry 5.0, a significant change is taking place, signifying a departure from Industry 4.0 and introducing a new period marked by the cooperative interaction between humans and machines (Turner et al., 2021). This progression adds revolutionary features such as digital twins and blockchain technology, which are set to completely transform the field of supply chain management. Digital twins are virtual replicas of physical systems that provide real-time analysis and monitoring. They are crucial for data-driven decision-making and optimizing operational efficiency. The concept of digital twins originated from the Apollo 13 mission and has undergone significant refinement over the years (Alzakri, 2023; Li & Gong, 2023). Simultaneously, blockchain technology plays a crucial role in strengthening the trustworthiness and consistency of supply chains by serving as a secure and transparent digital record that guarantees secure and unchangeable transactions (Chen et al., 2023). Industry 5.0 seeks to unite modern technology and human interaction, creating a collaborative workspace that combines the best aspects of both realms (Su et al., 2020). An essential aspect of this transformative process involves combining digital twin and blockchain technologies, which have the remarkable capability to guide supply chain management towards a future that is both sustainable and transparent (Li et al., 2022). As we further investigate these phenomena, it is crucial to thoroughly examine the various advantages and difficulties presented by these technologies to fully realize the potential of Industry 5.0 and its impact on the future of supply chain management.

In the context of Industry 5.0, digital twins play a crucial role in connecting the physical and digital domains. They create a virtual representation of the real environment, enabling real-time analysis and simulations to improve supply chain processes (Umar et al., 2021). Integrating cyber and physical components in manufacturing units enhances predictive maintenance and reduces resource wastage, leading to a transformative era of sustainable corporate operations (Akram et al., 2023). Concurrently, the advancement of blockchain technology surpasses its initial purpose of emerging as a symbol of openness and protection in several sectors (Lee et al., 2022). Blockchain's decentralization characteristic in supply chain management creates an unchangeable record, improving the capacity to track and hold different parts of the supply chain accountable (Bah et al., 2021). This innovative technology ensures the genuineness of transactions, providing a strong solution to combat counterfeiting and unethical behavior, consequently guiding enterprises toward sustainability and ethical practices (Bourcet, 2020).

Integrating digital twin and blockchain technology foresees a future supply chain framework characterized by continuous monitoring and validation of processes, creating an environment where data accuracy and operational effectiveness live peacefully (Sahin & Sahin, 2023). This symbiotic relationship is poised to push a transition towards digital twin technologies (DTT), placing resource optimization and waste reduction at the forefront and integrating industry operations with global sustainability goals (Liang et al., 2023). To navigate this disruptive period, it is necessary to examine real-world examples where combining digital twin and blockchain technologies brings advantages (Yin et al., 2023). Instances vary from automotive manufacturing facilities utilizing digital twins for virtual testing and component development to agriculture supply chains employing blockchain for product authenticity and traceability, consequently transforming industry norms (Kelishomi & Nisticò, 2022). These case studies provide strong evidence of the revolutionary capacity of these technologies, presenting a clear plan for achieving a more sustainable and efficient supply chain ecosystem. Nevertheless, this trip is not without its hurdles, as problems connected to data privacy, scalability, and interoperability emerge as significant barriers to the seamless application of these technologies (Jiglau et al., 2024). One example of a challenge related to data privacy in the integration of digital twin and blockchain technologies is evident in healthcare, where ensuring patient confidentiality while utilizing digital twins for treatment simulations is critical (Hemdan et al., 2023). However, integrating blockchain for secure data sharing among healthcare providers poses complexities in maintaining privacy. Solutions could include implementing privacy-preserving techniques such as homomorphic encryption or zero-knowledge proofs (Gökalp et al., 2018). Regarding scalability, consider the challenge faced in automotive manufacturing when deploying digital twins across multiple factories, straining computational resources. Solutions might involve optimizing algorithms or adopting distributed ledger technologies with higher transaction throughput. Similarly, scaling blockchain networks to handle increasing transaction volumes for product traceability in agriculture supply chains presents scalability hurdles that could be addressed through off-chain solutions or network optimizations (Sharma & Kumar Tiwari, 2023).

Moreover, incorporating digital replicas into current systems and the significant energy usage associated with blockchain operations highlight the necessity for a cautious and strategic approach to their implementation, underscoring the importance of regulatory guidance, industry partnerships, and ongoing research and development. To tackle these difficulties, a comprehensive strategy that promotes the growth of digital twin and blockchain technologies while upholding ethical and environmental norms was adopted (Figueira et al., 2023; Kirikkaleli et al., 2023). In the future, the combination of digital twin and blockchain technologies has the potential to bring about a new era in supply chain management. This period will be defined by the smooth integration of innovation and sustainability (Chen et al., 2023). As researchers and industry specialists explore the complexities of these technologies, a unique opportunity emerges to reshape the field of industry operations, bringing about a new era where technology accelerates sustainable growth and progress (Basu et al., 2022).

The current era has seen significant technological advancements that have strengthened the interdependence between humans and machines. It is crucial to recognize the vital role of digital twin and blockchain technologies in shaping a supply chain network that prioritizes sustainability and efficiency (Awosusi et al., 2023). This concept is centered around combining the physical and digital aspects of supply chain operations. It holds the potential for a future where predictive analytics and safe transactions become the standard (Chatterjee et al., 2023). The core element of this revolutionary journey is the robust framework of digital twin technology, which is increasingly acknowledged as a fundamental aspect in attaining efficient and environmentally friendly manufacturing processes (Jianqiang et al., 2020). Digital twins utilize real-time data analysis and simulation of real-world situations to provide exceptional predictive capabilities and operational management, significantly reducing downtime and improving product quality (Shen et al., 2022). This complex system, strengthened with sensors and analytics, enables a smooth exchange of information that can be utilized to make well-informed choices, thereby minimizing inefficiency and advocating for environmental sustainability (De Sousa et al., 2020). Simultaneously, blockchain technology is recognized as a symbol of transparency and security in supply chain management. It operates decentralized, eliminating the typical obstacles in centralized systems and facilitating a more efficient and transparent process flow (Abid et al., 2022). By implementing it, stakeholders can obtain unchangeable records of transactions, promoting an environment of trust and responsibility, which is crucial for establishing a durable and adaptable supply chain (Battisti et al., 2022).

Furthermore, thoroughly comprehending the complexities associated with digital twin and blockchain technologies highlights the need to foster partnerships between industry pioneers and policymakers, enabling the smooth incorporation of these technologies into conventional supply chain operations (Ashenafi & Dong, 2023). Focusing on this goal can improve operational efficiencies and contribute meaningfully to achieving global sustainability targets as defined in important agreements such as the Paris Agreement and the Sustainable Development Goals (Xu et al., 2023). This collaborative paradigm, based on a commitment to innovation and sustainability, has the potential to fundamentally change the way industries operate. It envisions a future where technology catalyzes positive change and sustainable growth (Ingendahl et al., 2021). The combination of digital twin and blockchain technology is expected to revolutionize supply chain management, creating a new approach emphasizing innovation and sustainability (Allal-Chérif et al., 2022). The upcoming journey requires a collaborative approach, where collective efforts and continuous research foster a landscape characterized by effectiveness, openness, and environmental responsibility. From the perspective of Industry 5.0, we may envision a future where technology and humans come together to create a more promising and environmentally friendly future.

## Literature Review

The combination of digital twin and blockchain technologies in the context of Industry 5.0 presents a significant opportunity to create a more sustainable future in supply chain management. An extensive examination of the current literature uncovers a thorough investigation of the possible advantages and consequences linked to the integration of these technologies. The deployment of digital twin technology in Industry 5.0 has received significant scrutiny, particularly its ability to provide a seamless connection between the physical and digital domains (Zhang et al., 2022). This technology enables continuous monitoring and proactive maintenance, reducing waste and optimizing resources. According to Milana and Wu (2012), digital twin technology catalyzes processes and makes informed decisions by creating a virtual representation of physical goods, which eventually promotes sustainable practices in supply chain management. Blockchain technology has been acknowledged for improving transparency and traceability in supply chain sustainability, as noted by Magnac and Roux (2021). The decentralized structure of blockchain guarantees the confidentiality and unchangeability of data, fostering an atmosphere of trust and cooperation among participants in the supply chain (Wang et al., 2022). This academic investigation highlights the significant capacity for combining digital twin and blockchain technologies in the framework of Industry 5.0, revealing a route towards a supply chain ecosystem that is more sustainable and resilient.

Furthermore, the amalgamation of blockchain and digital twin technologies improves the ability to trace and make transparent the flow of goods within the supply chain and enables the thorough monitoring of products from their production to their consumption. The aforementioned capability fosters responsible sourcing methods and is a powerful deterrent against fraudulent activities and counterfeit products, advancing the cause of sustainable and ethical industry operations (Hernández et al., 2022). An analysis of the convergence of these technologies reveals a growing collection of academic literature indicating their capacity to transform the sector, resulting in a supply chain marked by improved sustainability, efficiency, and transparency. Bigerna et al. (2023) contend that combining blockchain and digital twin technologies can lead to a significant revolution by enhancing the resilience and sustainability of supply chains through continuous monitoring and unchangeable record-keeping. Their proposed integrated system has the potential to bring about a new era of sustainability. It involves carefully tracking, monitoring, and optimizing products and processes in real time. This approach aims to minimize adverse environmental effects and encourage corporate responsibility, as mentioned by Bahl et al. (2020). This discussion showcases the latest research and scholarly viewpoints on the significant consequences of combining blockchain and digital twin technologies. It demonstrates the possibility for a fundamental change towards a more environmentally friendly and accountable supply chain system. Therefore, this paper proposes the following hypothesis:

> H1: Implementing digital twin technology positively influences sustainable supply chain management in Industry 5.0.

In the field of Industry 5.0, experts argue that combining digital twin and blockchain technology might be a crucial element in building a long-lasting and cooperative production system that focuses on human needs (Zhan et al., 2021). The integration is believed to mark the beginning of a period in which supply chains become more efficient, robust, and aligned with environmental and social sustainability objectives (Ma et al., 2023). Existing literature indicates that integrating digital twin and blockchain technologies within the Industry 5.0 framework can lead industries toward sustainable supply chain management characterized by increased transparency, efficiency, and social responsibility. Recent research suggests that combining digital twin and blockchain technologies will likely significantly change supply chain management. This convergence can redefine industry norms by promoting innovation and sustainability (Duong et al., 2022).

Upon closer examination of the inherent qualities of these technologies, it becomes evident that digital twin technology functions as a means for virtual testing and simulation, allowing for thorough analysis before physical production. This strategy saves resources and conforms to sustainability standards, reducing adverse environmental effects (Han et al., 2022). Moreover, blockchain technology's authentication and validation features enhance the establishment of a transparent and secure supply chain network. As emphasized by Moslehpour et al. (2022), the network has a crucial function in increasing customer confidence by offering unparalleled clarity on the source and environmental impact of products, thereby promoting a culture of conscientious consumption. The scholarly discourse focuses on cutting-edge research in Industry 5.0, highlighting the revolutionary power of combining digital twin and blockchain technology to reshape supply chain dynamics for a sustainable and socially responsible future. Based on this, the study proposes the subsequent hypothesis:

> H2: Implementing blockchain technology has a direct positive effect on sustainable supply chain management in Industry 5.0.

Furthermore, experts argue that the smooth incorporation of digital twin and blockchain technologies into supply chain management has the capacity to bring about a period marked by increased cooperation and effectiveness. In this setting, the crucial importance of stakeholders becomes vital. Vachon and Klassen (2007) argue that actively involving stakeholders at different points in the supply chain is essential for maximizing the potential advantages of these technologies. Moreover, the shift towards a more environmentally friendly supply chain requires a change in the culture and attitude of the business. The developing paradigm promotes the exchange of knowledge and collaborative endeavors, creating a work climate favorable to innovation and ongoing education. This method is crucial for successfully incorporating digital twin and blockchain technologies into the sector and effectively managing their difficulties (Toh et al., 2014). In the future, it is important to prioritize and focus on starting research and development projects that explore the combined potential of these technologies in a more comprehensive way. Developing frameworks that enable the seamless integration of digital twin and blockchain technologies is essential for developing a robust and sustainable supply chain management system in the upcoming era of Industry 5.0 (Verma et al., 2022). Equally important is the development of educational programs aimed at improving the expertise of industry personnel, equipping them for a smooth shift towards a technologically updated supply chain infrastructure. Facilitating collaborations between academia and industry can significantly contribute to generating innovation and promoting the wider implementation of these technologies (Bramwell et al., 2012). This academic discussion emphasizes the need for collaboration across many fields, ongoing education, and deliberate research efforts to effectively utilize the revolutionary capabilities of digital twin and blockchain technologies in the changing context of Industry 5.0. According to this, the study puts out the following hypothesis:

H3: Enhanced transparency in the supply chain positively mediates the relationship between implementing digital twin technology and sustainable supply chain management in Industry 5.0.

As we explore the concept of Industry 5.0, our main goal is to create a supply chain system that aligns with the larger goals of protecting the environment and being socially responsible, which set a new standard for an industry that emphasizes innovation, collaboration, and sustainability. The shared goal is to imagine a future in which supply chains demonstrate efficiency, agility, and unshakable commitment to ethical responsibility and environmental stewardship, which lead us into a new era of industrial evolution where sustainability plays a vital role. To successfully navigate future transitions, it is crucial to have a deep understanding of the potential implications and opportunities that arise from combining digital twin and blockchain technology. This understanding is built upon the insights gathered thus far. These technologies significantly alter supply chain efficiency and sustainability standards when effectively integrated, influencing the shift toward Industry 5.0 (Maddikunta et al., 2022). In this new and unique environment, the technology known as digital twin becomes a crucial tool for closely monitoring the whole lifespan of products. It provides valuable information that can be used to improve the design and production processes. Data-driven solutions enhance production efficiency and significantly contribute to waste reduction and resource optimization, driving the sector toward a more environmentally conscious and sustainable path (Liu et al., 2021). This discourse emphasizes the importance of developing a deep understanding of the complex relationship between digital twin and blockchain technologies to navigate the changing landscape of Industry 5.0. This understanding is crucial for achieving a sustainable and socially responsible future for the industrial sector, going beyond simple technological integration.

Simultaneously, blockchain technology constitutes a robust framework for fortifying data security and augmenting transparency within supply chain networks. Its inherent decentralization ensures the immutability and security of data, engendering a climate of trust and collaboration among diverse stakeholders (Findlay, 2017). This elevated level of transparency and security is anticipated to engender ethical practices, diminish instances of fraud, and cultivate a corporate culture wherein organizations are held accountable for their actions. Consequently, this dynamic is poised to nurture a more sustainable and responsible business ecosystem, aligning with the principles of corporate social responsibility and ethical conduct (D'amato et al., 2009). Therefore, this paper proposes the following hypothesis:

H4: Enhanced transparency in the supply chain positively mediates the relationship between implementing blockchain technology and sustainable supply chain management in Industry 5.0.

Beyond its immediate features, digital twin technology emerges as a transformational cornerstone in the changing Industry 5.0 ecosystem. As a pillar of this industrial revolution age, digital twin technology is not limited to resource optimization; rather, it is seen as a constructive intermediary in the relationship between supply chain management goals and implementation (Banholzer, 2022). The premise of this mediation is that the implementation of digital twins will improve operational effectiveness, bring about real-time monitoring capabilities, and enable well-informed decision-making processes in the context of supply chains (Botín-Sanabria et al., 2022).

The symbiotic integration of digital twin technologies is expected to cause Industry 5.0 to undergo a paradigm shift. Its pivotal function as an intermediary drives the sector toward a future in which the realization of sustainable supply chain practices depends on optimal resource utilization. Utilizing Digital Twin's ability to simulate, test, and analyze actual products in real time, the supply chain may attain previously unheard-of levels of efficiency while also cutting waste and allocating resources optimally (Tosini, 2020). This paradigm's emphasis on sustainability and environmental awareness places the digital twin in a position that makes it possible to create a supply chain ecosystem that adheres to Industry 5.0's core values. Thus, incorporating digital twin technology goes beyond traditional technological uses and acts as a catalyst to transform the fundamentals of industrial processes. This conception is based on encouraging a supply chain that is sustainable and environmentally conscious in addition to optimizing resource use. The potential influence of digital twin technology as Industry 5.0 develops goes beyond a simple technological improvement; rather, it is a strategic development that foresees a time when technology and sustainable practices harmoniously coexist at the center of industrial progress (Leng et al., 2022). Therefore, the following hypothesis is put out in this paper:

> H5: Optimized resource utilization resulting from digital twin technology positively mediates the relationship between implementing digital twin technology and sustainable supply chain management in Industry 5.0.

When considering Industry 5.0, the adoption of blockchain technology represents a paradigm change toward transparent and safe transactions. The best use of resources from using blockchain technology acts as a beneficial moderator in the interaction between it and sustainable supply chain management (Long et al., 2023). This theory is based on blockchain's decentralized and secure characteristics, which are expected to result in increased operational effectiveness, fewer instances of fraud, and increased transparency overall. Thus, these characteristics play a major role in creating a supply chain ecosystem that is socially and sustainably responsible (Seuring et al., 2008). The decentralized structure of blockchain guarantees transactions are safe, transparent, and impervious to manipulation, which fosters confidence among supply chain participants. This increased trust is anticipated to promote cooperation and moral behavior, which are crucial elements of an ethical and sustainable economic environment (Hosmer, 1995).

Additionally, the decentralized structure of blockchain provides extra protection for supply chain transactions by lowering the susceptibility to fraudulent activity. Given these characteristics, blockchain technology's ability to maximize resource use becomes apparent as a key intermediary in the transition of Industry 5.0 towards sustainable supply chain management methods (Leng et al., 2022). Due to its capacity to increase productivity, reduce fraud, and foster transparency, blockchain technology is viewed as a critical facilitator in transforming Industry 5.0 into a supply chain management era that is sustainable and morally grounded. This theoretical paradigm emphasizes that blockchain is important to Industry 5.0 as a technological tool and a catalyst for creating a supply chain ecosystem that complies with sustainability standards (Gigauri & Janjua, 2023). Therefore, the following hypothesis is put out in this paper:

> Hypothesis 6 (H6): Optimized resource utilization from blockchain technology positively mediates the relationship between implementing blockchain technology and sustainable supply chain management in Industry 5.0.

As shown in Fig. 1, the proposed research model shows how technology transforms supply chains, particularly blockchain technology. Blockchain, a decentralized ledger technology, promotes supply chain transparency by storing transactions securely and immutably. This technology will provide real-time visibility into the location and condition of items, from raw materials to completed goods, across the supply chain. Transparency throughout the supply chain is thought to improve operational efficiency and reduce waste. The concept also views blockchain technology as a mediator between digital twin technology and resource optimization. Supply chain simulation and optimization depend on digital twin technology, which simulates physical products and processes. The suggested model suggests blockchain technology can improve digital twin accuracy and dependability, maximizing resource use. The research model depicts blockchain technology as a key player in supply chain management, boosting transparency, operational efficiency, and resource utilization in Industry 5.0.

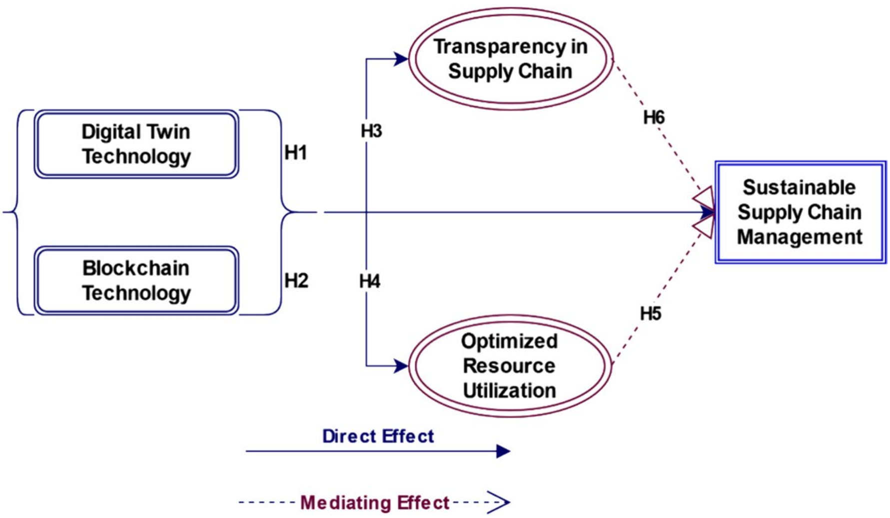


**Fig. 1** The proposed research model

The integration of DTT and blockchain technology (BCT) in the context of Industry 5.0 offers substantial potential for fostering sustainability in supply chain management. Existing literature underscores DTT's role in virtual testing and simulation, promoting resource optimization and environmental consciousness, while BCT enhances transparency, security, and ethical sourcing practices. This amalgamation is projected to revolutionize supply chain dynamics by ensuring traceability, mitigating fraud, and fostering cooperative ecosystems. Moreover, the strategic incorporation of DTT and BCT is essential for achieving sustainability goals in Industry 5.0, requiring stakeholder involvement, cultural shifts within organizations, and continuous research and development efforts. By synthesizing insights from previous studies, the current research aims to contribute to this evolving discourse, proposing hypotheses that investigate DTT and BCT's direct and mediating effects on sustainable supply chain management within the Industry 5.0 framework.

## Methodology

The research methodology employed a rigorous approach to assess the implementation of Industry 5.0 (I5.0) technologies in Bangladesh's Ready-Made Garments (RMG) industry. Utilizing a carefully constructed component model developed from industry insights and literature review, the study evaluated the willingness of industry managers to embrace I5.0 technologies. Criteria for assessment were derived from expert insights and influential research in sustainable performance evaluation, economic and social performance analysis, sustainable supply chain management (SSCM) activities indicators, and digital twin technology (DTT) practices. Data collection involved a comprehensive questionnaire, pilot testing, and analysis using partial least squares structural equation modeling (PLS-SEM) to validate the proposed hierarchical component model and hypotheses, ensuring methodological rigor and reliable conclusions regarding the adoption of Industry 5.0 (I5.0) technologies in the RMG sector.

### Measurements of the Construct

This study introduced a carefully constructed component model to thoroughly examine the implementation of I5.0 technologies, specifically focusing on the ready-made garments (RMG) industry. The model, as described in Zhang et al.'s (2022) research, was developed by combining insights from industry specialists and conducting a comprehensive assessment of relevant literature. It successfully identified the essential technological advances necessary for the implementation of Industry 5.0 in the RMG industry. The advancements included the implementation of machine vision to detect flaws in textiles, using RFID systems for managing inventory, the application of robotics for complex sewing operations, and utilizing digital twins for supply chain management. Although these technologies have been widely adopted in industrialized countries, their implementation in emerging economies, namely, in labor-intensive RMG enterprises in Bangladesh, has presented significant difficulties. This study aimed to evaluate the willingness of industry managers in Bangladesh's RMG organizations to embrace I5.0 technologies, considering the duration of their implementation in different enterprises.

The study utilized measuring criteria generated from expert insights to assess the importance of I5.0 technologies. The criteria used to evaluate a corporation's sustainable performance, including environmental factors, were derived from the influential research conducted by Shaw and Scully in 2023. In addition, the criteria used to analyze economic and social performance were based on academic research conducted by Ingendahl et al. (2021) and Xu et al. (2023). Allal-Chérif et al. (2022) identified indications and assessment criteria for SSCM activities through a study that incorporated feedback from industry professionals and a review of relevant literature. The formulation of aspects and evaluation criteria for DTT practices was achieved by combining industry expert assistance and literature sources (Abid et al., 2022; Ashenafi & Dong, 2023; Battisti et al., 2022). The study utilized a 5-point Likert scale to determine the perceived significance of adopting I5.0, applying SSCM techniques, and commencing DTT procedures. The scale ranged from 1 ("not necessary") to 5 ("urgent/mandatory"). This scale allowed for a subtle comprehension of participants' perspectives on the significance of different aspects within the three dimensions. To evaluate the performance of ORU, data was collected using a 5-point Likert scale. This data was obtained through a comprehensive questionnaire that covered all variables, survey components, and relevant sources.

## Pilot Testing of the Proposed Model

Before starting the extensive survey, a preliminary test was carried out to confirm the dependability and content validity of the survey tool. This procedural stage adhered to accepted methodologies as delineated by De Sousa et al. (2020) and entailed seeking input from academic experts and industry professionals to evaluate the soundness of the survey questions. Input was first gathered from eight academics with specialized knowledge in Industry 5.0 and supply chain. The specialists thoroughly evaluated the significance and clarity of the scale measures used in the survey. Afterward, in accordance with the technique suggested by Bertrand et al. (2022), a preliminary survey was conducted with 40 participants to assess the reliability of the questionnaire. The aim was to determine if the survey participants could accurately understand each question and if the items were expressed in a clear and concise manner. Respondents' feedback during the initial phase led to modifications in the wording of multiple questions to improve clarity and remove redundant or unclear topics. Concurrently, the gathered replies aided in evaluating the reliability and validity of the construct. Initial analysis utilizing partial least squares structural equation modeling (PLS-SEM) revealed that the survey findings exhibited reliability and validity levels that met acceptable criteria. Building on the favorable results of the pilot test, the study smoothly transitioned to surveying on a larger scale. The survey instrument was subjected to a careful process, which involved following known methodologies and continuously refining it based on input. This approach highlighted the methodological rigor to ensure the survey instrument's robustness and effectiveness.

## Analytical Approach

The PLS-SEM approach was used to quantitatively assess the proposed hierarchical component model and its associated hypotheses. In this context, a two-stage strategy was employed within the model to examine the hierarchical variable framework, yielding more reliable results compared to the repeated indicating method (Shen et al., 2022). At first, the measurement frameworks of both lower and upper-tier variables were examined closely. An analysis was conducted to evaluate the formative measurement approach and assess the products' reliability and validity (R&V), including convergent and discriminant aspects. This assessment employed the variance inflation factor (VIF) in conjunction with the examination of external weights and loadings. After confirming the accuracy of item R&V in both models' levels, the next step was to assess the structural model. The objective of this phase was to validate the provided hypotheses using bootstrapping processes with 5000 subsamples, following suitable criteria. The SmartPLS 3.3.9 program was used for this purpose. In addition, to confirm the suitability of the provided framework, calculations were made for the R-square values and Stone Geisser's Q2 as part of the validation process. Based on the PLS-SEM method, this rigorous analytical procedure follows the highest academic criteria, guaranteeing a thorough investigation of the hierarchical component model and hypotheses to derive dependable and legitimate conclusions.

## Mitigating the Risks to R&V of the Findings

While creating the questionnaire and conducting the research, great care was taken to detect and reduce any biases affecting the reliability and credibility of the study's results. To address the importance of these biases, proactive steps were taken to systematically incorporate preventive measures throughout the creation of the questionnaire and data collection. The objective was to guarantee impartial responses and maintain the dependability of the results. An in-depth comprehension of possible obstacles, such as the possibility of experts giving identical answers or respondents displaying "social desirability" behavior, was derived from previous research studies, including those conducted by Makhdoom et al. (2023) and Ahmed et al. (2022). The inquiry also considered the possible existence of common method bias (CMB), a factor that has been shown to compromise the dependability and accuracy of results, as demonstrated in studies conducted by Liu et al. (2023) and Makhdoom et al. (2023). The research employed thorough procedural tactics to tackle these difficulties, including careful planning and rigorous statistical analysis after collecting the data. To mitigate common method bias during data collection, we implemented proactive measures. These measures included segregating items that measured independent and dependent variables and introducing a time delay between responses for self-reliant and reliant factors. These actions were in accordance with the recommendations provided by Ahmed et al. (2022) and Han et al. (2022). In addition, the study formulated unambiguous questions that were concise and straightforward, including relevant examples as needed to reduce any potential confusion. The approach expanded to include statistical analysis conducted after data gathering. The study employed the PLS technique, as described by Duong et al. (2022), to examine the entire collinearity VIF of the latent variables. The comprehensive results are shown in Table 1, which shows the CMB test findings using internal VIF coefficients for important latent variables.

The VIF values, which indicate multicollinearity due to technique bias, are usually much below 5. Digital twin technology and blockchain technology have VIFs of 1.259 and 1.357, indicating low common method bias. Transparency in the supply chain has a VIF of 1.714, indicating a somewhat higher but acceptable bias. Optimized resource utilization has a low VIF of 1.089, indicating little method bias. Sustainable supply chain management's 1.500 VIF shows moderate but acceptable bias. While care is suggested, none of the variables exceed the crucial threshold, indicating that common technique bias does not affect responses. Researchers may take extra steps, notably for transparency in the supply chain and sustainable supply chain management, where VIF values are higher, to strengthen the findings and ensure study validity.

Our extensive investigation confirmed the absence of significant bias in our findings, as indicated by the VIFs being below 3.3, which aligns with prior research conducted by Ma et al. (2023) and Zhan et al. (2021), which also found no evidence of bias. In addition, the research also examined the possibility of non-response bias, which could have undermined the reliability and generalizability of the data, which was accomplished by employing a paired sample *t* test in SPSS, which compared the early responses (first 100 samples) with the following ones (the latter 100 samples) to identify any possible disparities in the feedback between these two groups, as recommended by Bahl et al. (2020) and Bigerna et al. (2023). According to Table 2, all the factors showed *p* values greater than 0.05, indicating no significant difference between the initial and subsequent groups of replies. Therefore, we can confidently state that our data was not affected by non-response bias.

Table 2 shows a detailed analysis of non-response bias using *T* tests on paired samples across different parameters, including digital twin technology early (DTT-E), DTT-L, blockchain technology early (BCT-E), BCT-L, transparency in the supply chain early (TSC-E), TSC-L, ORU-E, ORU-L, and Sustainab, and the significance levels, average scores, and standard deviations for each parameter in early and late response groups are given. The *p* values over 0.05 show that all parameters' early and late responses are similar, implying no significant non-response bias across evaluated parameters. The consistent responses of early and late respondents strengthen the study's findings and data validity. The *T* test results show no non-response bias in the dataset, verifying the study's findings and ensuring their credibility.

**Table 1** CMB test utilizing internal VIF coefficients

| | Inner VIF |
|---|---|
| Digital twin technology | 1.259 |
| Blockchain technology | 1.357 |
| Transparency in the supply chain | 1.714 |
| Optimized resource utilization | 1.089 |
| Sustainable supply chain management | 1.500 |

**Table 2** Evaluating non-response bias with *t* tests on paired samples

| Parameter | Resp. Cat | Samp. Count | Avg | Std. Dev | Sig. Level |
|---|---|---|---|---|---|
| DTT-E | Early | 105 | 4.1321 | 0.44518 | 0.968 |
| DTT-L | Late | 105 | 4.1382 | 0.44499 | - |
| BCT-E | Early | 105 | 4.0898 | 0.51578 | 0.931 |
| BCT-L | Late | 105 | 4.0821 | 0.51566 | - |
| TSC-E | Early | 105 | 4.0301 | 0.71352 | 0.900 |
| TSC-L | Late | 105 | 4.0438 | 0.69789 | - |
| ORU-E | Early | 105 | 4.1579 | 0.57345 | 0.741 |
| ORU-L | Late | 105 | 4.1873 | 0.58299 | - |
| SSCM-E | Early | 105 | 4.1549 | 0.56753 | 0.739 |
| SSCM-L | Late | 105 | 4.1842 | 0.57705 | - |

*DTT* digital twin technology, *BCT* blockchain technology, *TSC* transparency in the supply chain, *ORU* optimized resource utilization, *SSCM* sustainable supply chain management

## Results

A careful examination of the evaluation processes linked to the hierarchical element model's structural and measurement framework strictly followed accepted academic standards. Furthermore, the outcomes of the mediation evaluations were methodically defined, guaranteeing an academic and thorough exposition of the study results.

### Evaluating the Measurement Model

The initial assessment involved examining the formative measurement model by closely analyzing the outer weights of all variables. The variables "effective data management infrastructure," "Industrial IoT implementation," "reduced consumer complaints," and "environmentally friendly production processes" were eliminated since they did not fulfill the significance threshold for their outer weights. Following that, the focus was redirected towards assessing the psychometric properties of fundamental concepts essential for understanding the higher-level concepts in the model—specifically, the acceptance of Industry 5.0, sustainable supply chain management, digital twin technology practices, and operational resource utilization. Bootstrapping techniques were utilized to compute the VIF and *t* values to mitigate concerns regarding collinearity. Following the criteria specified by Hernández et al. (2022), all VIF values were below 5, and the *t* values above 1.96, demonstrating the lack of collinearity issues in the model. The model's robustness has positioned it for the next level of structural investigation.

### Assessing Multicollinearity Issues

Bootstrapping was used to evaluate multicollinearity by calculating the VIF values. The results of this examination are elaborated in Tables 3 and 4, which are. The

**Table 3** Outcome of hypothesis tests

| Hypothesis | Path description | Path coefficients | *t* value | *p* value | Outcome |
|---|---|---|---|---|---|
| H1 | DTT → SSCM in Industry 5.0 | 0.662 | 10.480 | 0.000 | Confirmed |
| H2 | BT → SSCM in Industry 5.0 | 0.755 | 15.820 | 0.000 | Confirmed |
| H3 | DTT → TSC → SSCM in Industry 5.0 | 0.732 | 16.130 | 0.000 | Confirmed |
| H4 | BT → TSC → SSCM in Industry 5.0 | 0.642 | 7.870 | 0.000 | Confirmed |
| H5 | DTT → ORU → SSCM in Industry 5.0 | 0.317 | 3.215 | 0.002 | Confirmed |
| H6 | BT → ORU → SSCM in Industry 5.0 | 0.373 | 3.635 | 0.000 | Confirmed |

**Table 4** Mediation analysis

| Hypotheses | Total impacts | | Direct impacts | | Indirect impact | |
|---|---|---|---|---|---|---|
| Description | Path Coeff: | *t* value | Path Coeff: | *t* value | Hypotheses | Path Coeff: |
| DTT → SSCM | 0.658 | 10.370 | −0.050 | 0.460 | DTT → TSC → SSCM | 0.228 |
| DTT → SSCM | 0.658 | 10.370 | −0.050 | 0.460 | DTT → ORU → SSCM | 0.479 |
| BT → SSCM | 0.594 | 9.345 | −0.49 | 0.543 | BT → TSC → SCM | 0.387 |
| BT → SSCM | 0.594 | 9.345 | −0.49 | 0.543 | BT → ORU → SSCM | 0.345 |

findings showed that the VIF values for all variables were below the specified threshold of 5. This observation suggests that there are no substantial problems of collinearity in the model, indicating that it is appropriate for further structural evaluation.

## Evaluating the Structural Models

During the initial phase of our study, the R-squared ($R^2$) values were computed for our hierarchical factor framework. $R^2$ values gauged the extent to which independent variables (INDV) could account for the variability in dependent variables (DV). In our study, the $R^2$ values for SSCM, DTT, and operational resource utilization (ORU) performance were determined to be 0.560, 0.526, and 0.709, respectively. To provide context to these results, it was imperative to note that, as per Wang et al. (2022), conventional $R^2$ values of 0.67 indicated a significant correlation, values of 0.33 suggested a moderate connection, and values of 0.19 indicated a weak connection. In the current investigation, the recorded values fell within the "moderate" category, which is considered to be within an appropriate range.

Our hypothesis tests on Industry 5.0 factors are shown in Table 3. The results support the first hypothesis (H1) that digital twin technology (DTT) and sustainable supply chain management (SSCM) in Industry 5.0 are positively correlated. The path coefficient is 0.662, and the *t* value is 10.480. The path coefficient of 0.755 with a *t* value of 15.820 supports the second hypothesis (H2) that blockchain technology (BT) and SSCM in Industry 5.0 are positively correlated. In H3 and H4, transparency in the supply chain (TSC)–mediated DTT/BT and SSCM. H3 and H4 had path coefficients of 0.732 and 0.642, respectively, with significant $t$ values, confirming TSC's mediation of these connections. H5 and H6, which hypothesized the mediation of ORU between DTT/BT and SSCM, were confirmed by path coefficients of 0.317 and 0.373, respectively, with significant $t$ values. In conclusion, all assumptions were confirmed, demonstrating the beneficial links between DTT, BT, TSC, ORU, and SSCM in Industry 5.0 and shedding light on these complex supply chain management relationships.

### Mediation Analysis

This study examined the complex mediation dynamics of SSCM and DTT practices in the context of Industry 5.0 (I5.0) technologies. It specifically investigated their influence on ORU efficiency, as outlined in hypothesis 6. The preliminary evaluation of H6 demonstrated a statistically significant and positive direct impact (H1) between I5.0 technologies and ORU efficiency, showing a substantial association ($\beta = 0.674$, $t = 16.370$, $p < 0.05$). However, when SSCM practices were introduced as a mediator, the previously established direct relationship became less significant ($\beta = -0.057$, $t = 0.464$). At the same time, the nonlinear impact of SSCM procedures continued to be statistically significant ($\beta = 0.245$, $t = 3.036$). This result is consistent with the attributes of either complete or competitive mediation, as demonstrated by a prior study conducted by Magnac and Roux in 2021. Regarding hypothesis 6, which examined the impact of DTT practices as a mediator, the analysis revealed a significant and favorable overall effect (H1) between I5.0 technologies and ORU efficiency ($\beta = 0.644$, $t = 11.360$, $p < 0.05$). However, when DTT practices were introduced as a mediator, the direct connection became less important ($\beta = -0.057$, $t = 0.464$). At the same time, the nonlinear effects related to DTT behaviors continued to be substantial ($\beta = 0.476$, $t = 6.304$), further indicating the need for comprehensive or competitive mediation. Ultimately, both H6 and H7 received statistical validation, highlighting the role of SSCM and DTT practices in positively influencing the connection between I5.0 technologies and supply chain sustainability performance.

Table 4 shows the mediation analyses' critical evaluation and interpretation of the hypotheses' effects. The mediation study examined the total, direct, and indirect influences of defined routes. The first pair of hypotheses, DTT → SSCM and BT → SSCM, had total impacts of 0.658 ($t = 10.370$) and 0.594 ($t = 9.345$), respectively. However, both pathways had negative coefficients and were statistically insignificant, demonstrating that DTT or BT did not directly affect SSCM. DTT and BT had indirect effects of 0.228 and 0.387, respectively, on SSCM, mediated by TSC. Additionally, the mediation paths DTT → ORU → SSCM and BT → ORU → SSCM showed significant indirect influences of 0.479 and 0.345, respectively, indicating that ORU strongly mediated the effects of DTT and BT. These results demonstrate the complex mediation dynamics within the proposed pathways, emphasizing the roles of supply chain transparency and optimized resource utilization in mediating digital twin technology, blockchain technology, and sustainable supply chain management.

## Correlation Analysis

When examining the relationship between hidden variables, we followed the guidelines proposed by Milana and Wu (2012) and Gong et al. (2023). We utilized the partial least squares (PLS) technique to assess latent variables, analyze their correlations, and detect significant correlations, regarded as values surpassing 0.9. Identifying significant connections between hidden variables was essential as it could suggest the possible existence of common method bias (CMB). Table 5 displays correlation data showing all identified correlations below the essential threshold of 0.9. This finding provided reassurance that the presence of common method bias, caused by confounding variables, did not significantly impact the validity of our study. The thorough assessment confirmed the strong methodology of our investigation and increased the trust in the dependability of our results.

The association matrix for our latent variables—digital twin (DT), blockchain (BT), transparency (TC), resource utilization (ORU), and sustainable supply chain management (SSCM)—is shown in Table 5. These connections are essential for understanding major construct links. Correlation coefficients show relationship strength and direction. The correlation between DT and BC is 0.688, indicating a moderately favorable association. TC, DT (0.664), and BC (0.622) have considerable positive associations, highlighting their interdependence. ORU's connection with DT (0.616), BC (0.712), and TC (0.515) shows the intricate relationship between optimum resource usage and other technological components. Sustainable supply chain management's correlations with DT (0.693), BC (0.724), TC (0.634), and ORU (0.729) demonstrate its broad links to Industry 5.0 technologies. Various correlation data help clarify the interplay between dimensions, directing debates on the supply chain ecosystem's overall impact of multiple technologies.

The data revealed that none of the constructs had a correlation value over 0.9. Therefore, it was confidently stated that the existence of CMV did not present a significant problem in the study, in accordance with the guidelines outlined by Zhang et al. (2022) and Shaw and Scully (2023). Common method variance, a potential confounding factor in survey-based studies, arises when respondents' views of constructs are influenced by a shared method, resulting in artificially inflated correlations. To limit the risk of common technique bias and improve the reliability and validity of the research findings, the study ensured that the correlation values among

**Table 5** Correlation amongst hidden variables

| Variable | DT | BT | TC | ORU | SSCM |
|---|---|---|---|---|---|
| DT (digital twin) | 1 | | | | |
| BC (blockchain) | 0.688 | 1 | | | |
| TC (transparency) | 0.664 | 0.622 | 1 | | |
| ORU (resource utilization) | 0.616 | 0.712 | 0.515 | 1 | |
| SSCM (sustainable SCM) | 0.693 | 0.724 | 0.634 | 0.729 | 1 |

*DT* digital twin technology, *BC* blockchain technology, *TC* transparency in the supply chain, *ORU* optimized resource utilization, *SSCM* sustainable supply chain management the constructs were kept below the recommended level of 0.9. The thorough analysis of correlation values enhanced the reliability of the methodology and bolstered the trustworthiness of the study's findings.

## Effects of CVs in the Framework

When assessing the possible interference of control variables (CVs) on our research framework, we examined how professionals' gender and years of professional experience affected the overall framework and the results of our hypothesis analysis. After carefully analyzing  it became clear that the inclusion of CVs did not substantially impact the endogenous variables. The implementation of CVs resulted in a little elevation of the beta values for all the postulated pathways. Nevertheless, it is imperative to recognize that neither gender nor the duration of job experience significantly influenced the efficacy of optimized resource utilization (ORU). The research revealed that the correlation between gender and ORU performance resulted in a *t* value of 1.395, indicating a modest effect size. The *p* value of 0.081 suggested that this association did not reach statistical significance at the customary threshold of 0.05. Likewise, the correlation between the number of years of job experience and ORU performance yielded a *t* value of 0.928, suggesting a small magnitude of impact. The *p* value of 0.177 indicated that this association was not statistically significant. Hence, we can confidently state that the control variables had no substantial impact on our suggested architecture.

## Model Validations

To evaluate the strength of our model, we utilized Stone-Geisser's Q2 value, which was calculated using a blindfolding procedure. According to the established technique, it is typically anticipated that Q2 values will be positive, indicating trustworthy predictive skills and well-designed dependent variables (DVs). The study revealed Q2 values of 0.223, 0.270, and 0.492 for the implementation of SSCM practices, digital twin technology (DTT) techniques, and the performance of ORU, respectively.

## Test of Bias Result from Endogeneity

To guarantee the precision and reliability of our results, we thoroughly examined the issue of endogeneity, which is a situation that emerges when certain variables are not included or when there are violations of the homogeneity requirement in the partial least squares-structural equation modeling (PLS-SEM) model. Endogeneity adds bias in coefficient estimations and can result in erroneous causal inferences, particularly when reciprocal linkages exist between endogenous and exogenous factors. Researchers have suggested different strategies to tackle endogeneity, such as instrumental variable approaches and instrument-free techniques like joint estimation using Gaussian copulas. Our study utilized the SmartPLS Gaussian copula technique, following the guidelines suggested by Allal-Chérif et al. (2022) to address the issue of endogeneity. The findings, indicated that all Gaussian copulas yielded statistically insignificant outcomes, with $p$ values surpassing 0.05. In addition, our examination of copula structures inside the PLS model revealed no statistically significant findings ($p > 0.05$). Hence, we confidently affirmed that our inquiry did not uncover any endogeneity in accordance with the findings reported by Ingendahl et al. (2021).

## Discussion

In the ever-changing environment of Industry 5.0, the smooth incorporation of digital twin (DT) and blockchain technology signifies a significant change in perspective towards attaining SSCM (SHARKAWY, 2020). Our study thoroughly examines the complex interactions between these new technologies and how they affect the efficient use of resources in the supply chain system. Initial results confirm a strong connection between Industry 5.0 technologies and the effectiveness of ORU, providing convincing proof for the deliberate implementation of technology to improve overall performance measures. Nevertheless, the narrative becomes more intricate when we delve into the intermediary functions of SSCM and dynamic capability (DT) practices (Mathivathanan et al., 2017). Using these practices as mediators, the formerly significant direct connection between Industry 5.0 technologies and ORU performance weakens, suggesting the possibility of complete or competitive mediation. This perceptive observation highlights the complex interaction of various elements within the supply chain, underlining the necessity for a sophisticated strategy in implementing technology adoption initiatives (Birasnav & Bienstock, 2019). In addition, the analysis of possible confounding variables (CVs) and the application of Stone-Geisser's Q2 value enhance the model's stability and predictive ability, strengthening our research findings' credibility. By utilizing SmartPLS Gaussian copula techniques to address endogeneity concerns, statistically insignificant results confirm the lack of endogeneity (Hult et al., 2018), which emphasizes the significant impact of DT and blockchain technologies in reshaping Industry 5.0 and promoting sustainable and optimized supply chain management practices.

Moreover, the study methodically tackled the possibility of non-response bias by utilizing techniques firmly rooted in the direction of earlier landmark research. This thorough examination not only guaranteed the reliability of our data but also strengthened them against the dangers of significant bias, solidifying the validity of our conclusions. The rigorous methodology consistently demonstrated no significant difference in variance between the first and subsequent groups of responses, confirming that the data was accessible from non-response bias. Furthermore, the study conducted a thorough analysis of the relationship between latent variables, following the strict guidelines established by earlier researchers, and employed the partial least squares method to avoid common method bias and ensure the integrity of the research as a whole. In addition, a detailed examination was carried out to investigate any confounding influences of control variables, strengthening the investigation against biases and confirming the resilience of the research framework. Given these results, the study not only confirms the critical role that digital twin technology (DTT) and SSCM practices play in mediating the complex relationship between Industry 5.0 (I5.0) technologies and optimized resource utilization (ORU) performance, but it also makes the case that these practices must be fully integrated and adopted in order to fully realize the benefits that I5.0 technologies have to offer (Singh et al., 2023). The study highlights how these technologies have the potential to revolutionize supply chain dynamics in a way that promotes efficiency, sustainability, and transparency. This research, which is positioned as a lighthouse in the digitalized industry landscape, is evidence of the great potential at the nexus of blockchain and digital twin technologies. It points toward a future where supply chain management in Industry 5.0 is more resilient, efficient, and sustainable. As such, it invites more research into this exciting field in an effort to fully understand the complex web of advantages and implications it has for contemporary supply chain dynamics.

In the dynamic landscape of Industry 5.0, the integration of DTT and blockchain technology (BCT) represents a pivotal shift towards SSCM. Through meticulous examination, our study elucidates the intricate interactions between these technologies and their impact on ORU performance (Maghrabi, 2014). While initial findings underscore the direct influence of Industry 5.0 technologies on ORU, the intermediary functions of SSCM practices and dynamic capabilities (DT) present a nuanced perspective. Concrete examples reveal how SSCM practices, such as transparency initiatives and ethical sourcing methods, facilitate the efficient utilization of resources by enhancing visibility and accountability throughout the supply chain. Similarly, dynamic capabilities empower organizations to adapt and innovate in response to technological advancements, optimizing resource allocation and fostering sustainability (Mele et al., 2024). By uncovering these mediating mechanisms, our research underscores the critical importance of fully integrating DTT and BCT alongside SSCM practices and dynamic capabilities to realize the transformative potential of Industry 5.0 technologies in promoting efficiency, sustainability, and transparency in supply chain management.

To build a sustainable supply chain management (SSCM) framework, this study clarifies the relationships and importance of Industry 5.0 technologies, with an emphasis on DTT and blockchain technology (BCT). Empirical research suggests that integrating these technologies has the potential to yield significant gains in resource utilization (ORU), which is a crucial component in promoting sustainable industrial practices. The research demonstrates the considerable effect of SSCM and DTT techniques on improving ORU performance by thoroughly examining their mediating effects. Beyond scientific results, the study has important policy ramifications that will help the industry innovate and become more transparent and sustainable. Acknowledging important limitations like the study's limited reach and reliance on quantitative data, further research topics are highlighted. It is suggested that mixed-method study designs be used, that a wider range of parameters be investigated, and that longitudinal studies be carried out to evaluate how these technologies change over time.

The integration of DTT and BCT brings forth intricate data privacy and ethical implications. Firstly, while DTT enhances data collection and analysis for improved decision-making, it raises concerns about data privacy due to the extensive amount of personal and sensitive information collected and stored (Varriale et al., 2023). Ensuring the security and confidentiality of this data becomes paramount to prevent unauthorized access or misuse. Additionally, BCT, known for its immutable and transparent ledger, presents ethical dilemmas regarding data ownership and transparency. While it enhances trust and accountability by providing a tamper-proof record of transactions, it also challenges traditional notions of privacy by making transactional data permanently accessible to all network participants (Braz et al., 2008). Thus, reconciling the benefits of transparency with individuals' right to privacy becomes a critical ethical consideration in the integration of DTT and BCT. Striking a balance between data transparency and privacy protection is essential to mitigate ethical concerns and foster responsible use of these technologies.

Furthermore, qualitative studies exploring the experiences of industry personnel could provide complex viewpoints. The paper promotes the incorporation of state-of-the-art technology to mold an inventive, productive, and sustainable future for the industrial sector, acting as a lighthouse in the ever-changing Industry 5.0 environment. It validates the industry's upward trend and creates opportunities for innovation and ongoing development. The study's closing thoughts, which look forward to a sustainable future and express hope, predict that the information acquired would spur previously unheard-of growth and sustainability in the industrial sector. The study's theoretical and policy implications highlight how well it advances current knowledge, offers a solid theoretical framework, and advises policymakers on encouraging technological adoption and prioritizing sustainability in Industry 5.0.

## Conclusion

This research comprehensively examines the complex relationship between Industry 5.0 technologies, specifically DTT and BCT, and their crucial role in promoting a SSCM framework. The empirical evidence strongly confirms the transformative ability to incorporate these advanced technologies, particularly in enhancing the efficient use of resources, which is a crucial factor in attaining sustainability in industrial operations. The study provides evidence for the mediating effects of SSCM and demand and transportation time practices, demonstrating their significant impact on improving the effectiveness of ORU performance. The study has numerous ramifications, providing a diverse range of valuable information that can inform policy development and direct the industry toward a path marked by innovation, transparency, and sustainability. In the industrial sector, the integration of technologies such as DTT and BCT in supply chain processes is not just an innovation but a must. This integration is essential for creating an efficient and sustainable future. While recognizing the significant contributions of the study, it is important to note its limits and provide directions for future research, such as broadening the scope, including a wider variety of variables, and using more sophisticated methodology. This study serves as a prominent example in the ever-changing realm of Industry 5.0, providing theoretical and policy implications that highlight its significance in improving existing knowledge and directing knowledgeable policymakers toward increasing technological adoption and prioritizing sustainability. The paper ends by envisioning a hopeful, sustainable future through the transformative power of incorporating modern technology into supply chain operations. This integration is expected to promote ongoing improvement, innovation, and remarkable growth in the industrial sector.

The results of this study provide a significant and remarkable contribution to the theoretical discussion in the emerging topic of Industry 5.0. The paper explores the complex relationship between DTT, BCT, and SSCM. It sheds light on a previously overlooked connection in the theoretical field. The study offers a detailed insight into how the incorporation of Industry 5.0 technologies, notably DTT and BCT, impacts and improves the efficient use of resources (ORU) in the context of the supply chain. This intricate contribution is crucial for theoretical progress in multiple ways. Primarily, it tackles significant deficiencies in the current body of knowledge by elucidating the complexities of the connections between these fundamental elements of Industry 5.0. The paper presents a comprehensive theoretical basis that effectively combines technology elements with sustainability considerations, offering a more comprehensive framework for scholars and researchers to investigate. This enhanced theoretical comprehension helps clarify the mechanics involved in Industry 5.0 and provides opportunities for future research investigations and model improvements. Researchers can utilize these theoretical understandings as a starting point for additional investigation, which may result in more advanced models that accurately depict the intricate interactions within the intricate terrain of Industry 5.0. This study's theoretical contributions enhance our understanding of the complex dynamics that influence the future of industrial processes. It offers valuable insights that go beyond the specific focus of the research.

The findings obtained from this study can significantly influence the development and modification of policies, especially within the complex context of Industry 5.0. The presence of compelling data about the beneficial effects and intermediary functions of sustainable supply chain management (SSCM) and digital twin technology (DTT) practices provides a strong basis for policymakers to develop well-informed and future-oriented policies. Within the framework of Industry 5.0, these policies can deliberately prioritize and encourage the extensive incorporation of disruptive technologies, such as DTT and BCT, into supply chain activities. The primary objective would be to foster a conducive atmosphere for innovation, improve supply chain transparency, and advance sustainability measures. Policymakers may steer the industry towards a future marked by increased efficiency, ongoing innovation, and a strong dedication to sustainability by prioritizing these crucial issues. This report presents itself as a significant and timely resource for policymakers dealing with the intricate aspects of Industry 5.0. It provides evidence-based insights that align with the changing dynamics of the industrial sector and significantly contribute to the formulation of policies that promote the smooth integration of advanced technology and sustainable practices. As policymakers confront the complexities and possibilities brought about by sector 5.0, the results of this study serve as a guiding light, offering a clear plan for making smart policy choices that can direct the industry towards a forward-thinking and environmentally friendly path.

Although this study is the first to investigate the complex relationship between digital twin and blockchain technology for sustainable supply chain management in Industry 5.0, it is important to recognize its inherent constraints. The research primarily utilized quantitative data, a methodological decision recognized for its strength and reliability; nonetheless, it may not fully capture the intricate dynamics and subjective experiences that qualitative insights can offer. Moreover, the study intentionally limited its focus to particular aspects of Industry 5.0, disregarding other potentially significant factors and technologies that were not investigated. Although variance inflation factors (VIFs) and a paired sample *t* test were used with great attention to detail, it is recognized that these procedures may not fully capture all possible biases, which suggests the necessity for continuous improvement in methodological approaches. To enhance the thoroughness of future investigations, it would be beneficial to include a wider range of variables and adopt mixed-method research designs to obtain more significant insights. Longitudinal studies could be conducted to track the progressive effects and advantages of various technologies as they develop over time.

Furthermore, qualitative research that explores the direct experiences and viewpoints of professionals in the industry shows the potential to provide a more comprehensive comprehension of the practical consequences and obstacles related to the integration of digital twin and blockchain technology in the supply chain. By adopting these approaches, future research efforts can cultivate a comprehensive and enhanced comprehension of the diverse effects of these revolutionary technologies within the framework of Industry 5.0, which will expand the limits of existing knowledge and promote advancements in sustainable supply chain management.

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**Data Availability** The datasets and materials used and/or analyzed during the current study are available from the corresponding authors on reasonable request.

## Declarations

**Conflict of interest** The authors declare no competing interests.

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